

ANKIT KUMAR

Delhi, India

+91 7320087900 ankitkumar220520041@gmail.com linkedin.com/in/ankit-kumar-5a447627b/
github.com/ankitkumar-22

EDUCATION

- **Birla Institute of Technology, Mesra** 2026
B.Tech in Electronics and Communication Engineering CGPA: 8.29
- **Narbheram Hansraj English School** 2022
Indian School Certificate (ISC), New Delhi Percentage: 95
- **Hill Top School** 2020
Indian Certificate of Secondary Education (ICSE), New Delhi Percentage: 94.2

EXPERIENCE

- **Samsung Research & Development, Noida** Jan 2026 - Jun 2026
SDE Intern Noida
 - Built a **Flask-ADB** diagnostic utility adding **illegal IMEI detection** at the customer service tier, removing engineering-escalation dependency and cutting diagnosis time to under 60 seconds; in staged rollout to **300+ Samsung service centers** nationwide.
 - Rebuilt an internal file-transfer server (**Django, Nginx**) that hard-failed above 30 MB, implementing **chunked multipart uploads** with client-side segmentation, **async server-side reassembly**, and SQL-backed chunk-state tracking; removed the size cap, enabled **resumable uploads**, deployed across **5 teams serving 200+ users**.
 - Designed a Carrier Aggregation extraction engine using **PyQt6 and Pandas** indexing 500+ config entries across 10+ Qualcomm spec files into a unified searchable interface, cutting engineer lookup time by **95%+** (5+ min to **under 10 sec**).
- **Tata Motors** May 2025 - Jul 2025
SDE Intern Jamshedpur
 - Built a fault-isolated data pipeline replacing twice-weekly manual transfers with 15-minute aggregation cycles across **11 assembly-line sensor systems (165+ Excel reports per run)**; engineered per-source fault isolation for file locks, network drops, and machine-offline events, ensuring **zero-downtime** continuity.
 - Built a real-time safety-compliance system (**YOLOv10, Python**) processing **5 live CCTV feeds at 10 FPS** on standard laptop hardware, with automated **helmet-violation detection** and bounding-box audit logging, achieving **94%+ accuracy** on Bay 11 of the production line.

PROJECTS

- **Distributed Network Monitoring System**  Jun 2026
Tools & technologies : Java, Spring Boot, Apache Kafka, MongoDB, ShedLock, REST API, Docker
 - Built a horizontally scalable, **fault-tolerant** distributed system with independent scheduler, worker, and ingestion roles (one **Spring Boot** app via profiles) over partitioned **Apache Kafka** topics, scaling out as **consumer groups**.
 - Guaranteed single-active scheduling via a **MongoDB**-backed distributed lock (**ShedLock**) and **ordered, idempotent** processing through device-ID partition keying; containerized with **Docker**.
- **Inventory Management Microservices Backend**  Jun 2025
Tools & technologies : Spring Boot, MongoDB, OpenFeign, REST API, Kafka, Java MailSender, Docker
 - Built a **Spring Boot** backend of 5 independent microservices (Product, Order, Supplier, Notification, Inventory) communicating via REST APIs with **OpenFeign, Eureka** service discovery for dynamic routing, and **MongoDB** for persistence.
 - Added **Kafka** async messaging between Order and Notification services, a **Spring Cloud Gateway** as unified API entry point, and **Docker Compose** for single-command deployment.
- **Predictive Maintenance (RUL Prediction)**  Feb 2025
Tools & technologies: Python, TensorFlow, XGBoost, Pandas, Random Forest, Scikit-learn
 - Built a **RUL prediction** system on **NASA C-MAPSS (FD001)**, removing 7 redundant sensors via Pearson correlation with RUL clipping and 10-cycle rolling-mean augmentation, achieving **RMSE 32.48** (R2 0.71) with **SVR** and 4-fold cross-validation.
 - Extended it to a **3-tier risk classifier** (No Risk, Moderate, Critical) using **SVC** and **Random Forest** with stratified splits, reaching **68%** validation accuracy across imbalanced classes.

TECHNICAL SKILLS

Languages: Java, Python, SQL, Bash

DevOps: Docker, Kubernetes, Git, GitHub Actions (CI/CD), Nginx

Backend: Spring Boot, REST APIs, Microservices, Kafka, OpenFeign, Django, Flask

Databases: MySQL, MongoDB

AI / ML: LangChain, Groq API, SQLAlchemy, Scikit-learn, XGBoost, SVR, Random Forest, YOLOv10, Plotly/Dash

Tools & Concepts: Git, ADB, TCP/IP, HTTP/HTTPS, OSI Model

ACHIEVEMENTS

- **Leetcode Knight** (rating 1900+) Jun 2026
- **Graduate Aptitude Test in Engineering (GATE) in DA** Secured AIR 1388 Feb 2025
- **Joint Secretary, Robolution Robotics Club** Feb 2024